11. Extreme Programming
Embracing Change

"Software Development Methodologies"
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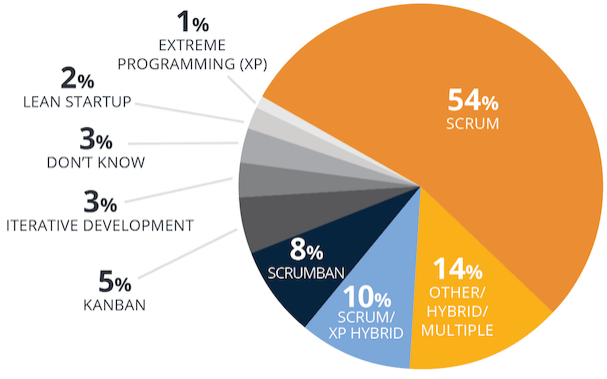
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Extreme Programming

- Lightweight incremental software development process.
- Involves a high-degree of discipline from the development team
- Popularized by K. Beck (late 90's)
  - 1st edition 1999
  - 2nd edition 2004
- Comprises 12 core practices
- Rapid feedback from customer
- Assumes simplicity – simple design
- Incremental change – small changes

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Extreme Programming

- According to R.Mall and R.Fojtik XP mostly applies to:
  - Projects with new technologies or research projects
  - Projects with small dimensions

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XP Values

- Communication
  - Between programmers through pair programming, daily stand-up meeting, planning game
  - With customer on-site, and involvement in writing acceptance tests and the planning game
- Simplicity
  - Do the simplest thing that could possibly work
  - Use simple index cards, avoid speculative design for possible change, avoid generalized components that aren't immediately justified, …
- Feedback
  - Drives quality and adaptation
  - Test-first development, continuous integration, daily tracker, short iterations, frequent releases, …
- Courage
  - Develop fast, change fast!
- Respect
  - The whole team commits to the goals and is accountable for the result
  - Trust and respect between the team members, management and the customers

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XP rules (1)

- Planning
  - User stories are written.
  - Release planning creates the release schedule.
  - Make frequent small releases.
  - The project is divided into iterations.
  - Iteration planning starts each iteration.
- Managing
  - Give the team a dedicated open workspace.
  - Set a sustainable pace.
  - A stand-up meeting starts each day.
  - The project velocity is measured.
  - Move people around.
  - Fix XP when it breaks.

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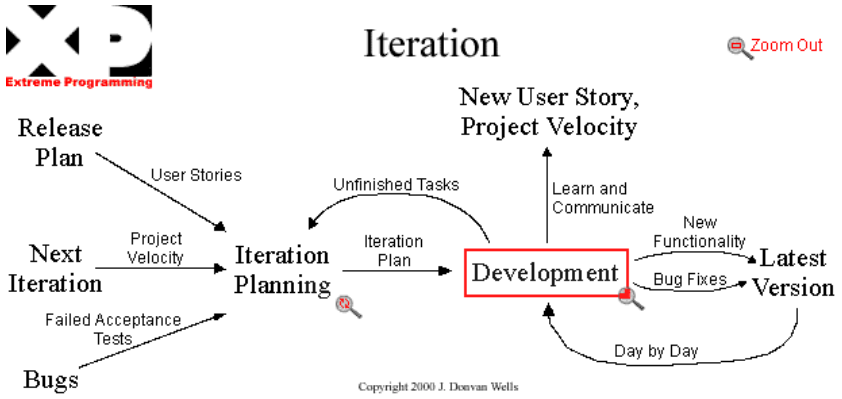
XP rules (2)

- Designing
  - Simplicity.
  - Choose a system metaphor.
  - Use CRC cards for design sessions.
  - Create spike solutions to reduce risk.
  - No functionality is added early.
  - Refactor whenever and wherever possible.
- Coding
  - The customer is always available.
  - Code must be written to agreed standards.
  - Code the unit test first.

XP rules (3)

- … Coding
  - All production code is pair programmed.
  - Only one pair integrates code at a time.
  - Integrate often.
  - Set up a dedicated integration computer.
  - Use collective ownership.
- Testing
  - All code must have unit tests.
  - All code must pass all unit tests before it can be released.
  - When a bug is found tests are created.
  - Acceptance tests are run often and the score is published.

One Iteration



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